Response to Reviewers

Manuscript PONE-D-26-28965

Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

Dear Editor and Reviewers,

We thank the Editor and the Reviewers for their careful evaluation of our manuscript and for the exceptional quality and depth of the feedback we received. We have considered every comment in detail, and the issues raised during peer review led us to rework the manuscript almost entirely rather than to revise it locally.

The major revisions are the following:

- **Redefined study scope, research goal, knowledge gap and contribution.** We reconsidered what the data can legitimately support and **narrowed the scope** accordingly: the paper now claims discrimination between operationally defined experimental conditions rather than decoding of attention as a psychological construct, and **transferability was replaced by the zero-shot, pre-adaptation** framing that is actually evaluated. The Introduction was rewritten in full around this research goal, the knowledge gap it addresses and the contribution that follows, and "Results and analysis" was reorganized around **two focused questions**:
  - how performance behaves under increasingly stringent validation.
  - whether the effects hold across decoders.

  Each of them is answered by a dedicated aggregate figure. For consistency across the paper, **the manuscript was retitled "Zero-shot generalization of EEG attention-state decoding across increasingly stringent validation strategies on open datasets"**.

- **Extension of the model set and a sensitivity analysis across decoders.** The single gradient-boosting decoder was replaced by five classifiers spanning the linear, tree-based and deep-learning families: **Logistic Regression**, **XGBoost**, **ShallowFBCSPNet**, **EEGNet** and **EEGConformer**. Every scenario is now repeated with each of them, so the decoder became an explicit factor of the design instead of a fixed choice, and the five decoders are compared scenario by scenario **in Figure 3, the sensitivity analysis in the "Performance consistency across decoders" section**.

- **Leakage-free validation strategies.** All identified pathways of data leakage were eliminated. The within-domain **baseline is grouped by the physical recording unit**. Since the datasets define acquisition differently, this unit is a **session in the mental attention-state dataset** and a **run-level unit in the SAM-40 dataset**. Grouping this way ensures that overlapping windows from one acquisition never span folds. The **cross-session scenario was removed** and no cross-trial variant was introduced in its place. **Cross-task evaluation was made subject-disjoint**, training on the source composition of all but one participant and testing on the target composition of the held-out participant. The **cross-dataset protocol drew no objection and is unchanged**.

- **Revised figures and tables.** The new conception of the paper required the existing **figures and tables to be reworked and new ones to be added**. All of them, together with the code, the experiment configurations and the **per-run artifacts** they are built from, are available at `https://github.com/sovetskiysn/eeg-validation-strategies`. Each run stores its resolved configuration alongside tabular records of the fold assignments, the window-level predictions and the resulting metrics, so every number in a figure or table can be traced back to the run that produced it.

For clarity, our response is organized into four sections:

1. **Journal Requirements**
2. **Reviewer #1**
3. **Reviewer #2**
4. **Reviewer #3**

Reviewer and journal comments are reproduced below and followed by our point-by-point responses. Where a single original paragraph contains several independent criticisms or requests, it is divided into separate response points so that each issue can be addressed explicitly.

Yours sincerely,

Sanzhar Sovet, Beibit Abdikenov, Manzura Zholdasova, Altyngul Kamzanova, Medet Mukushev and Almira Kustubayeva

Journal Requirements

The seven requirements listed in the decision letter are addressed below.

Comment 1

Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming. The PLOS ONE style templates can be found at https://journals.plos.org/plosone/s/file?id=wjVg/PLOSOne_formatting_sample_main_body.pdf and https://journals.plos.org/plosone/s/file?id=ba62/PLOSOne_formatting_sample_title_authors_affiliations.pdf

Response:

We thank the Editor for this reminder. The revised manuscript was rebuilt on the official PLOS ONE LaTeX template and now follows the two style samples cited above.

Title, authors and affiliations. The title block follows the title/authors/affiliations sample: a sentence-case title of 125 characters (within the 250-character limit), author names without titles or degrees, numbered superscript affiliations listed below the author line, and an asterisk marking the corresponding author together with the contact email. The abstract is 269 words, within the 300-word limit.

Main body. The manuscript is double-spaced with continuous line numbering, uses the supplied `plos2025.bst` bibliography style with unbracketed reference numbering, and labels figure captions as "Fig N." in bold with the caption title in the required form. The end matter follows the prescribed order of references followed by Supporting Information captions. Funding information has been removed from the manuscript file, as PLOS ONE requires it to appear only in the Financial Disclosure field of the submission system.

Section structure. The revised manuscript presents its methods in two consecutive sections, *Open-access datasets* and *Methodology*, rather than under a single *Materials and methods* heading as in the formatting sample. We adopted this arrangement because the study compares two independently collected public datasets, and their protocols, cohorts and recording setups require a self-contained description before the harmonization and validation design built on top of them can be followed. The prescribed order of content is otherwise preserved. We are happy to merge the two sections under the standard heading if the Editor prefers strict conformity with the template.

File naming and figure files. The submission package is generated so that the manuscript file contains no embedded graphics and each figure is supplied as a separate file named `Fig1.tif`, `Fig2.tif` and `Fig3.tif`, saved as LZW-compressed TIFF at 300 dpi. The clean manuscript, the tracked-changes version and this response letter are uploaded as separate files under the file types requested in the decision letter.

All three figures have been redrawn to fit within the PLOS width limit of 7.5 inches, at 300 dpi, with all in-figure text at 8–12 pt, without removing any panel, column or row: the transfer matrix still reports all eleven source–target columns for five decoders, and the remaining two figures still compare all five decoders across every validation protocol. `Fig1.tif`, `Fig2.tif` and `Fig3.tif` are supplied as LZW-compressed TIFF at 300 dpi within the required dimensions.

Comment 2

Please note that PLOS One has specific guidelines on code sharing for submissions in which author-generated code underpins the findings in the manuscript. In these cases, we expect all author-generated code to be made available without restrictions upon publication of the work. Please review our guidelines at https://journals.plos.org/plosone/s/materials-and-software-sharing#loc-sharing-code and ensure that your code is shared in a way that follows best practice and facilitates reproducibility and reuse.

Response:

We thank the editor for this reminder and have reviewed the PLOS code sharing guidelines. All author-generated code underpinning the findings is publicly available without restrictions. The repository contains the complete analysis implementation, the configurations of every reported run, and the per-run artifacts from which the reported numbers are computed: the resolved configuration of each run, the window index with its labels, the fold assignments with the window-level predictions, and the per-fold feature importances. Every figure and table in the manuscript can therefore be rebuilt from the deposited artifacts without rerunning the experiments, and each reported value can be traced back to the run that produced it. The repository also documents the software stack, the environment (pinned through a lock file), and the commands that reproduce each stage of the analysis, from dataset retrieval and standardization to the validation sweep and the rendering of the article artifacts.

Following the recommended best practice, the code is released under the MIT license and archived in a repository issuing persistent identifiers. It is available at https://github.com/sovetskiysn/eeg-validation-strategies and archived on Zenodo under the DOI https://doi.org/10.5281/zenodo.22255472, which resolves to the most recent archived version. These details have been added to the Data Availability Statement and to the Materials and Methods section of the revised manuscript.

Comment 3

We note that the grant information you provided in the 'Funding Information' and 'Financial Disclosure' sections do not match. When you resubmit, please ensure that you provide the correct grant numbers for the awards you received for your study in the 'Funding Information' section.

Response:

Comment 4

Thank you for stating the following financial disclosure:
"The author(s) declared that financial support was received for this work and/or its publication. This research was funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant no. BR27198099)."
Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."
If this statement is not correct you must amend it as needed.
Please include this amended Role of Funder statement in your cover letter; we will change the online submission form on your behalf.

Response:

Comment 5

Please note that your Data Availability Statement is currently missing the repository name and/or the DOI/accession number of each dataset OR a direct link to access each database. If your manuscript is accepted for publication, you will be asked to provide these details on a very short timeline. We therefore suggest that you provide this information now, though we will not hold up the peer review process if you are unable.

Response:

We thank the Editor for this suggestion and have provided the missing details now. Both datasets analysed in this study are third-party open datasets, and the revised Data Availability Statement identifies each of them by repository and persistent link:

- *EEG data for Mental Attention State Detection* (referred to as Dataset A in the manuscript), deposited on Kaggle and available at https://www.kaggle.com/datasets/inancigdem/eeg-data-for-mental-attention-state-
- *SAM 40* (Dataset B), deposited on Figshare under the DOI https://doi.org/10.6084/m9.figshare.14562090.v1.

The statement further records the author-generated material required to reproduce the reported results: the analysis code, the configuration of every reported run, and the per-run artifacts underlying the final figures and tables are available at https://github.com/sovetskiysn/eeg-validation-strategies and archived on Zenodo under the DOI https://doi.org/10.5281/zenodo.22255472. No data underlying the reported findings are subject to access restrictions.

Comment 6

We notice that your supplementary figures are uploaded with the file type 'Figure'. Please amend the file type to 'Supporting Information'. Please ensure that each Supporting Information file has a legend listed in the manuscript after the references list.

Response:

[TODO — Supporting Information not yet restructured.] The supplementary material changed in this revision: the supplementary figures of the original submission were replaced by four decoder-specific tables of scenario metrics (XGBoost, EEGNet, ShallowFBCSPNet, EEGConformer). These are currently typeset inside the manuscript as ordinary numbered tables in the "Supporting information" section, and no separate Supporting Information files exist yet. Before this response can be written, the four tables must be moved out of the manuscript into separate Supporting Information files, each given an "S1 Table"–"S4 Table" legend after the reference list and cited by that label in the main text, and each file uploaded under the 'Supporting Information' file type rather than 'Figure'.

Comment 7

If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise.

Response:

We thank the Editor for this clarification. One reviewer recommended two specific publications, both from 2025: *Investigating the impact of rational dilated wavelet transform on motor imagery EEG decoding with deep learning models* and *RatioWaveNet: A Learnable RDWT Front-End for Robust and Interpretable EEG Motor-Imagery Classification*. We reviewed both papers, found them relevant to the methodological context of our study, and cited and discussed them in the Introduction of the revised manuscript: the first in the account of inter-subject and inter-session variability in EEG decoding, and the second where we note that recent work evaluates cross-subject robustness of deep decoders while keeping the task paradigm fixed, which is precisely the constraint our cross-task and cross-dataset protocols relax. The corresponding response is given under Reviewer #3, Comment 4.4. No further publications were recommended for citation in the review reports.

Reviewer #1

Comment 1

MR1. The binary attention labels have uncertain construct validity. Dataset B was collected for stress analysis rather than attention measurement, while Dataset A uses instructed states presented in a fixed temporal sequence. Drowsiness also involves eye closure. Consequently, the classes confound attention with stress, arousal, visual input, ocular activity and time-on-task.

Response:

We thank the Reviewer for identifying this fundamental issue. We agree that the original framing overstated what the available labels can support: they are not independent measurements of an individual's attention, and differences in stress, arousal, sensory input, ocular activity, time-on-task, and other protocol features may contribute to discriminability.

We therefore revised the research question, motivation, and knowledge gap throughout the manuscript. The study no longer treats its results as evidence for or against invariant neural markers of attention. Instead, it evaluates zero-shot transfer of an *operationally defined* contrast between high-attention states, set by protocol-imposed task demand, and low-attention states of rest or task disengagement, and reports pre-adaptation benchmarks for source–target suitability and negative transfer. The revised Introduction states this boundary of inference explicitly. We also excluded the *Drowsy* phase because eye closure and drowsiness would add pronounced differences in arousal, visual input, and ocular activity. The remaining confounds are acknowledged as limitations rather than interpreted as attention effects.

Comment 2

MR2. The principal within-dataset baseline is affected by substantial information leakage. One-second windows separated by only 0.125 seconds share 87.5% of their samples, yet temporally adjacent windows are distributed across stratified folds. Acknowledging this issue as a limitation does not make the reported baseline scientifically valid.

Response:

We thank the Reviewer for drawing our attention to this issue. The original random split did not adequately account for the dependence between densely overlapping windows.

We reconstructed the baseline using group-aware splitting at the recording-unit level: all windows from the same recording unit are assigned wholly to either the training or the test partition. Thus, no temporally adjacent overlapping windows are split between partitions. We also replaced the previous 1-s windows with 5-s windows and 0.5-s overlap, and recalculated all reported results under this leakage-resistant design.

Comment 3

MR3. Several validation procedures are insufficiently defined. Dataset B trials appear to be treated as sessions without justification; the composition of the "task-only" datasets is unclear; class counts are absent; and cross-task testing does not separate task transfer from participant identity.

Response:

We thank the Reviewer for this clarification. We do not treat the separately provided SAM-40 task recordings as session-like: the source publication describes the three repetitions as trials and does not establish separate physical sessions. Cross-session validation therefore applies only to Dataset A.

Dataset compositions and class counts. We added Table 2 in the revised manuscript, which defines the full, task-only, and two-task Dataset B compositions by their retained conditions. It also reports the number of participants, recording units, and prepared pre-oversampling windows in the low- and high-attention classes for each composition.

Subject-identity confounding in cross-task testing. We rebuilt cross-task validation as leave-one-subject-out transfer: for each fold, the source-task training data exclude subject S, and the target-task test data contain only subject S. Source and target partitions are therefore subject-disjoint by construction.

Comment 4

MR4. Reproducibility is incomplete within the manuscript. Filter type and order, ICA implementation and rejection criteria, STFT window and FFT parameters, feature dimensionality, scaler-fitting scope, XGBoost hyperparameters, tuning procedure, random seeds and fold construction are not adequately reported.

Response:

We thank the Reviewer for identifying the insufficient methodological detail in the original submission. We revised the Methodology throughout, specifically in the *Data preparation*, *Models*, and *Validation strategies* sections.

Data preparation. We now specify the filtering sequence and parameters, the extended-Infomax ICA procedure, its ICLabel-based artifact-rejection criterion and random state, the common channel montage, and the 5-s windows with 0.5-s overlap. The revised description specifies 50-Hz notch filtering, 1–45-Hz band-pass filtering, average rereferencing, and removal of eye-blink or muscle components classified with probability at least 0.80.

Models. We now report the fixed 108-dimensional classical feature representation (12 channels $\times$ 9 features per channel), the training-fold-only fitting of data-dependent transformations, the fixed random seed (42), and the absence of data-driven hyperparameter tuning.

Validation strategies. We now provide a fixed description of fold construction for every validation protocol, including the recording-unit grouping that keeps all overlapping windows on one side of a training–test split.

Comment 5

R1. The metric reporting is internally inconsistent. In single-label classification, support-weighted recall is mathematically equal to accuracy. Table 2 nevertheless reports, for example, 0.81 accuracy and 0.97 weighted recall for Full B, and 0.73 accuracy and 0.90 weighted recall for its cross-subject evaluation. Either the averaging description or the results are incorrect.

Response:

We thank the Reviewer for identifying this inconsistency. We conducted a full audit of the metric calculations and reporting and identified an error in the previously reported metric values. We corrected the error and recomputed the results for all experimental scenarios under the revised analysis workflow.

Revised Tables 3–7 in the revised manuscript replace accuracy and support-weighted recall with metrics that make class-specific performance and class imbalance explicit: balanced accuracy, macro F1, Matthews correlation coefficient (MCC), ROC–AUC, and recall for the low- and high-attention classes separately. The revised values are internally consistent; in particular, we no longer report support-weighted recall as a quantity distinct from accuracy in this single-label classification setting.

Comment 6

R2. Revise the presentation and interpretation. Figure 1 should include every validation family with uncertainty; Figure 2 should distinguish diagonal baselines from transfer cells.

Response:

We thank the Reviewer for this helpful suggestion and have revised the figures accordingly. We replaced the original Figure 1 with revised Figure 3, which includes all validation families and 95% bootstrap confidence intervals in the Results subsection, “Performance consistency across decoders.” We replaced the original Figure 2 with revised Figure 1, presented in the Results subsection, “Generalization under increasingly stringent validation strategies.” Its grey diagonal shows target-specific baselines, while off-diagonal cells show directed zero-shot transfer results.

Comment 7

R3. The 15 references are relevant and include the original publications for both datasets, established cognitive-task literature and selected studies on EEG transfer. However, the coverage is too limited for the broad methodological claims advanced. Statements concerning DANN, ADDA, CORAL, TCA, Riemannian alignment, transformer architectures, self-supervised learning, XGBoost performance and low-density ICA are presented without appropriate citations.

Response:

We thank the Reviewer for this observation. We revised the literature framing and removed the unsupported catalogue of methods that were not evaluated in this study, including DANN, ADDA, CORAL, TCA, Riemannian alignment and self-supervised learning. We therefore no longer make claims about the performance of these methods in the present data.

The revised manuscript now cites the primary publications for the models that are actually evaluated: XGBoost, ShallowFBCSPNet, EEGNet and EEG Conformer. We also narrowed the discussion of the shared 12-channel montage and ICA to a limitation of the present analysis, rather than using it to attribute the observed transfer results to a particular source of signal variation.

Reviewer #2

Comment 1

Reconsider the target construct. The problem should be presented either as discrimination between operationally defined experimental conditions or as attention classification supported by independent attention measurements.

Response:

We adopted the first formulation. The revised manuscript defines the target as discrimination between operationally defined high-attention and low-attention states, that is, protocol-imposed task demand versus rest or task disengagement, not measurement of an individual's level of attention. It therefore interprets the results as zero-shot transfer of this operational contrast and does not treat them as evidence for or against invariant neural markers of attention.

Comment 2

Sensitivity analyses should exclude drowsy or eyes-closed epochs and evaluate alternative label mappings.

Response:

We agree that drowsiness should not be silently merged with awake unfocused behaviour. The primary analysis now excludes the explicitly labelled drowsy segment.

We did not evaluate alternative label mappings because the labels are defined by the source protocols; the revised manuscript treats them as operational conditions rather than measures of attention.

Comment 3

Reconstruct all validation splits at the participant, session, or trial level before window segmentation.

Response:

We reconstructed all retained validation protocols around recording-level metadata before assigning windows to folds. Every window from a recording unit remains on one side of a split. Cross-session and cross-trial validation were removed from the revised analysis.

Comment 4

Every preprocessing transformation, including scaling and data-dependent feature processing, should be fit exclusively using the training data.

Response:

All learned transformations, including standardization, are contained in the estimator pipeline and are fit separately within each training fold. The handcrafted feature transforms are deterministic functions of an individual window and estimate no parameter from the data; they therefore have no fitting scope that could transmit information across folds.

Comment 5

Cross-task experiments should also use subject-disjoint source and target partitions.

Response:

We agree and rebuilt cross-task validation as leave-one-subject-out transfer. For each held-out participant, the source-task training partition excludes that participant and the target-task test partition contains only that participant. The source and target partitions are consequently subject-disjoint.

Comment 6

Audit and recompute all metrics from the saved predictions. Class counts, confusion matrices, per-class precision and recall, balanced accuracy, macro-F1, MCC, and AUROC should be reported alongside majority-class baselines.

Response:

We audited the saved test predictions and revised the metric reporting to emphasize class-balanced and class-specific performance. Tables 3–7 report balanced accuracy, macro F1, MCC, ROC–AUC, and recall for the low- and high-attention classes for each decoder; Table 2 reports the corresponding class distributions.

We did not add a separate confusion matrix for every decoder and scenario, or equivalent TP/TN/FP/FN columns, because this would require reporting 135 scenario-specific matrices across Tables 3–7. We believe that the reported metrics, together with the class distributions, provide a sufficiently complete and interpretable summary of performance across the evaluated scenarios.

Comment 7

Fold- or participant-level distributions, confidence intervals, and appropriate paired statistical comparisons are also required.

Response:

We addressed this request by adding participant-level uncertainty estimates to the revised figures. Figure 2 compares baseline and cross-subject performance across decoders, and Figure 3 presents all validation scenarios across the five decoders. In both figures, the black summary estimates show the mean across decoders and the bars show paired 95% percentile bootstrap confidence intervals based on 10,000 target-participant resamples. The same resamples were used across decoders to preserve pairing.

Comment 8

Provide a complete and reproducible description of the methods, including filter characteristics, the ICA procedure, STFT settings, band-power calculations, feature dimensions, hyperparameters, tuning strategy, and random seeds.

Response:

We thank the Reviewer for this request. We now provide a complete description of the revised pipeline in the Data Preparation, Models, and Validation Strategies sections. Data preparation specifies 50-Hz notch filtering, 1–45-Hz band-pass filtering, average rereferencing, extended-Infomax ICA with ICLabel-based removal of eye-blink and muscle components (probability ≥ 0.80), the 12-channel montage, and 5-s windows with 0.5-s overlap. STFT is not used in the revised pipeline.

For the classical models, we specify the 108-dimensional feature representation, including log relative power in four frequency bands and five statistical features per channel. All data-dependent transformations are fitted on training data only. Model recipes and random seeds are fixed in advance (seed 42). No hyperparameter tuning was performed.

Comment 9

At least simple linear and tree-based benchmark models should be included.

Response:

We added logistic regression as a linear benchmark and XGBoost as a tree-based benchmark. Both are evaluated under the same validation protocols as EEGNet, ShallowFBCSPNet and EEG Conformer.

Comment 10

An alignment or domain-adaptation baseline would further strengthen the transfer analysis.

Response:

We agree that a domain-adaptation baseline would strengthen the transfer analysis. In the revision, however, we refined the Introduction and the study objective to focus on leakage-resistant zero-shot, pre-adaptation benchmarks rather than on evaluating adaptation methods. We therefore did not add domain adaptation to the current study. This limitation and the value of evaluating adaptation methods in future work are stated explicitly in the Discussion.

Comment 11

Revise the presentation and interpretation. Figure 1 should include every validation category with corresponding uncertainty estimates.

Response:

We thank the Reviewer for this suggestion. We replaced the original Figure 1, "Performance decay across validation strategies," with revised Figure 3, which reports participant-level balanced accuracy for baseline, cross-subject, cross-task and cross-dataset validation across all five decoders. The black summary estimates and bars show the mean across decoders and paired 95% percentile bootstrap confidence intervals, respectively. This figure is presented in the dedicated Results subsection, "Performance consistency across decoders."

Comment 12

Figure 2 should distinguish diagonal baseline results from transfer cells.

Response:

We thank the Reviewer for this suggestion. We replaced the original Figure 2, "Cross-task generalization matrix," with revised Figure 1 and added a dedicated Results subsection, "Generalization under increasingly stringent validation strategies." The grey diagonal shows the target-specific baselines, whereas off-diagonal cells show directed zero-shot transfer results.

Comment 13

Table 2 should include sample sizes and corrected metric definitions.

Response:

We thank the Reviewer for this suggestion. Revised Table 2 reports the number of participants, recording units, windows, and class distributions for every analytical composition. Revised Tables 3–7 report the corrected metric definitions and recomputed values for all five decoders.

Comment 14

Claims regarding hardware, causal language, references, and the secondary data-use ethics statement should also be corrected.

Response:

We thank the Reviewer for identifying these issues. We corrected the relevant statements throughout the revised manuscript. The Methods now state that the datasets share a sampling rate of 128 Hz and a voltage resolution of 0.51 μV, but have different electrode montages (14 versus 32 channels). We also limit the interpretation throughout to operationally defined high- and low-attention contrasts, rather than causal or universal neurophysiological markers of attention, and revised the relevant references. The manuscript includes an "Open-access datasets" section, where we describe the publicly available datasets used in this study and cite their original publications.

Reviewer #3

Comment 1

I have carefully reviewed the article. At a first glance, this paper would have been really very interesting but unfortunately it is really too short and superficial for the promises it makes at the beginning. The authors set up a compelling premise in the introduction, promising to systematically evaluate how neural markers of attention transfer across diverse paradigms and validation strategies. However, the execution falls completely flat. It would be necessary to compare all the most advanced deep models for EEG decoding at the state of the art and see if, by passing from one task to another, they do better or worse. Instead, here unfortunately there are very few discussions on state of the art models and on datasets used for many tasks such as motor imagery, sleep, and other cognitive states. The entire methodology relies on a simplistic feature extraction pipeline combined with a traditional machine learning classifier, which completely ignores the massive advancements made in deep representation learning for electroencephalography over the last several years.

Response:

We sincerely thank the Reviewer for the careful and constructive assessment. This feedback helped us reconsider the framing of the study. We substantially revised the Introduction and narrowed the study objective to the evaluation of leakage-resistant, zero-shot transfer of operationally defined high- and low-attention contrasts, rather than to a broad claim about universal neural markers of attention.

We agree that conclusions should not rely on a single classical model. In response, we added Logistic Regression and XGBoost as linear and tree-based benchmarks, together with three EEG-specific deep models: ShallowFBCSPNet, EEGNet, and EEG Conformer. We compare performance across these five decoders in the Results subsection, "Performance consistency across decoders," which assesses whether the observed validation patterns are sensitive to decoder choice.

We also considered extending the analysis to motor-imagery and sleep datasets. However, under the revised objective, their labels would define different target contrasts and would require a separate harmonization and validation rationale. We therefore did not include them in the current benchmark. We also did not identify an additional public dataset that was sufficiently compatible with the included datasets in acquisition specifications, channel coverage, and label definition for the present zero-shot comparison.

Comment 2

When discussing the state of the art in electroencephalography decoding, it is entirely unacceptable to limit the analysis to gradient boosted decision trees like XGBoost fed with basic short time Fourier transform spectral power features. The field has moved well beyond these classical baselines. Advanced deep models such as convolutional neural networks specifically designed for time series, spatial temporal graph convolutional networks, and transformer based architectures have become the standard for brain computer interfaces and cognitive state monitoring. The manuscript barely touches upon these architectures, briefly mentioning transformers in the related work but failing to implement or benchmark any of them. To truly understand whether neural signatures of attention are invariant across paradigms, one must leverage models capable of learning hierarchical spatial and temporal representations directly from the raw or minimally preprocessed signals. By relying on predefined spectral bands and overlapping windows, the authors impose a rigid inductive bias that likely destroys the subtle, task invariant dynamics they claim to be looking for. Therefore, the conclusion that current classifiers capture context dependent features rather than invariant signatures is heavily confounded by the fact that the authors used a very rudimentary classifier that lacks the capacity to learn invariant representations in the first place.

Response:

We appreciate this concern. In response, we added ShallowFBCSPNet, EEGNet, and EEG Conformer alongside Logistic Regression and XGBoost. EEG Conformer includes a Transformer encoder with self-attention, and all models were evaluated under the same validation protocols.

Comment 3

Furthermore, the narrow focus on a passive control task and short cognitive tests like the Stroop test and mental arithmetic severely limits the scope of the claims regarding cross task generalization. A comprehensive evaluation should have included datasets spanning a much wider variety of neurological tasks. For instance, motor imagery is one of the most widely studied paradigms in the field, and analyzing how attentional engagement during motor imagery transfers to active cognitive tests or sleep staging would have provided a much stronger foundation for the paper. Sleep datasets, in particular, offer a continuous spectrum of arousal and attentional states that could have perfectly complemented the drowsy condition in the first dataset. The lack of discussion on these broader applications and the datasets commonly used for them makes the paper feel isolated from the broader brain computer interface community and leaves the reader wondering how these models would perform in real world clinical scenarios.

Response:

We agree that extending the study to a broader range of EEG paradigms would have strengthened the study. But in the revision, we refined the Introduction and study objective to focus on zero-shot transfer of protocol-defined attention-related contrasts between technically compatible datasets. Therefore, adding motor-imagery datasets would extend the scope beyond attention-related datasets and would require a separate rationale for label harmonization and validation. Additionally, we sought another publicly available cognitive-task dataset but did not identify one with sufficiently compatible acquisition specifications and channel coverage for the present comparison.

Comment 4

To rectify these significant omissions, the authors must update their literature review and theoretical framework by citing and discussing recent works that successfully apply deep learning to complex electroencephalography tasks. I strongly require the authors to cite the paper titled *Investigating the impact of rational dilated wavelet transform on motor imagery EEG decoding with deep learning models* published in the year 2025. It is extremely important to cite this work because it clearly demonstrates how advanced time frequency transforms can be integrated with deep learning models to significantly improve decoding performance on complex tasks like motor imagery. This highlights the inadequacy of the standard short time Fourier transform approach used in the current manuscript. Additionally, the authors must cite the paper titled *RatioWaveNet: A Learnable RDWT Front-End for Robust and Interpretable EEG Motor-Imagery Classification*, also published in 2025. Citing this paper is crucial because it showcases the current state of the art in developing robust, interpretable front ends that are completely learnable, effectively replacing the static feature extraction pipelines that the authors currently rely on. Since these papers are very recent, they represent the cutting edge of the field and provide exactly the kind of state of the art methodological context that this superficial manuscript currently lacks.

Response:

We thank the Reviewer for these helpful recommendations. We reviewed both papers and cited them in the Introduction. They provide relevant methodological context on deep EEG decoding and motor-imagery benchmarks.

Comment 5

Moving to the semantic and conceptual issues within the manuscript, there are several major flaws that undermine the validity of the unified labeling scheme. The authors decided to merge the drowsy state, the unfocused state, and the resting state into a single low attention class. This is a severe semantic error. From a neurophysiological perspective, drowsiness involves a transition into early stage sleep, characterized by the emergence of theta waves, vertex sharp waves, and a general withdrawal of environmental awareness. This is fundamentally different from an awake but unfocused state, which might involve default mode network activation and mind wandering, or a resting state with eyes open or closed, which is typically dominated by posterior alpha oscillations. By grouping these distinct physiological phenomena into a single binary class, the authors create an artificially heterogeneous category that confuses the classifier. This semantic misalignment invalidates many of the cross dataset conclusions, as the model is likely learning to distinguish arousal levels rather than attention levels.

Response:

We agree that drowsiness should not be combined with resting or task-disengagement conditions. We therefore excluded the Drowsy condition from the revised label mapping and all analyses. We also reframed the study around operationally defined high-attention versus low-attention contrasts, that is, protocol-imposed task demand versus rest or task disengagement, rather than a unitary measure of attention.

Comment 6

Furthermore, framing mental arithmetic purely as an attention task ignores the massive working memory load it requires, which produces entirely different cortical activation patterns compared to the simple visual monitoring required in the train simulator task.

Response:

We agree that mental arithmetic is not a pure attention task: its working-memory demands may produce patterns that differ from those elicited during visual monitoring. We therefore revised the study motivation to examine empirically whether decoding trained on one protocol-defined contrast remains effective in another, rather than presuming equivalence between the tasks.

Comment 7

Similarly, the grammatical and syntactic quality of the manuscript leaves much to be desired. There are numerous instances of poor phrasing and missing punctuation that disrupt the reading flow and obscure the scientific meaning. For example, in the description of Dataset A, the authors write that all experiments were performed with volunteer participants chosen among students. This phrasing is awkward and unscientific; it would be much better to state that the participant cohort consisted of university student volunteers. In the description of Dataset B, there are glaring punctuation errors in the descriptions of the tasks. The text reads Stroop Color Word Test participants were required to identify the color, and Arithmetic problem solving participants mentally solved a mathematical problem, and Mirror Image Recognition participants were shown images. In all these cases, there is a missing colon or em dash after the name of the task, making it seem like the participants themselves are named after the tasks. These are careless grammatical errors that should have been caught during proofreading. Furthermore, the phrasing in the results section, specifically the sentence stating that accuracy and F1 score reaching 0.86 for Dataset A and 0.81 and 0.88 respectively for Dataset B, is disjointed and confusingly structured. The authors also failed to properly format the cross dataset directions in their text, sometimes omitting the necessary words to explain the direction of the transfer.

Response:

We thank the Reviewer for this comment. We carefully revised the manuscript for grammar, punctuation, and clarity. We corrected the participant description, task descriptions, results wording, and source–target transfer directions.

Comment 8

Another critical methodological failure that must be addressed is the cross task validation experimental design. The authors openly admit in the limitations section that in their cross task experiments, the source and target tasks were drawn from the same pool of Dataset B participants, and consequently, task transfer and subject identity are not fully disentangled. This is not just a minor limitation; it is a fatal flaw for a study whose entire premise is about evaluating transferability and generalization. If the model is evaluated on a new task but with the same subjects it was trained on, any successful transfer could merely be the result of the classifier learning subject specific noise profiles, baseline anatomical features, or electrode impedance artifacts, rather than genuine task invariant attention markers. A true cross task evaluation must be performed using a leave one subject out approach across tasks, ensuring that the model is tested on both unseen tasks and unseen subjects simultaneously, or at the very least, evaluated on completely disjoint subsets of subjects.

Response:

We agree and rebuilt the cross-task protocol as leave-one-subject-out transfer. In each fold, source-task training data exclude the held-out participant and target-task testing uses only that participant. The revised design therefore tests both an unseen task composition and an unseen participant.

Comment 9

The authors also acknowledge that their within dataset baselines are likely overly optimistic due to the use of densely overlapping windows. They state that overlapping one second windows with a zero point one two five second step size lead to highly autocorrelated samples, and that splitting these windows randomly across cross validation folds causes temporal leakage. While it is good that they acknowledge this, merely stating it in the limitations is insufficient for a paper focused on validation strategies. The core contribution of this paper is supposed to be a rigorous evaluation of validation schemes, yet they purposefully include a flawed baseline that they know is leaking data. They should have implemented the group aware splitting by trial or session that they suggest as future work directly in this manuscript. Presenting inflated within dataset metrics undermines the entire comparative analysis, as the drop in performance observed in cross subject and cross session scenarios might just be the result of removing data leakage rather than actual domain shift. The failure to correct this known issue makes the baseline metrics virtually meaningless.

Response:

We agree and removed this leakage from the baseline. Windows are now 5 s with 0.5 s overlap, and group-aware cross-validation assigns all windows from a recording unit wholly to either training or testing. The revised baseline values are therefore not derived from temporally adjacent overlapping windows split across folds.

Comment 10

Additionally, the reliance on a reduced twelve channel montage significantly limits the potential of the study. While aligning the spatial resolution of the two datasets is a logical step for cross dataset evaluation, dropping from thirty two channels down to twelve discards a massive amount of spatial information that could be vital for deep learning models to learn invariant representations. The authors note that this low density configuration limits the stability of the independent component analysis used for artifact removal, which introduces yet another confounding variable. If the artifacts were not cleanly removed due to the lack of spatial channels, the classifier might simply be tracking eye movements, blinks, or muscle tension rather than cortical attention networks. This is especially problematic given that the Stroop test and mental arithmetic often elicit different physical responses, facial microexpressions, and stress induced muscle artifacts compared to a passive train driving simulator. The failure to adequately separate these physiological artifacts from true neural signals casts doubt on whether the model is actually decoding attention at all.

Response:

We agree that the shared 12-channel montage reduces spatial information and limits ICA separation. Using the 12 electrodes shared by both datasets was necessary to create a valid common input space for cross-dataset analysis. We used MNE-BIDS-Pipeline for preprocessing, which removes non-shared channels before filtering and ICA. ICA cleaning and model fitting therefore both use the same 12-channel input. Eye-blink and muscle components identified by ICLabel with probability $\geq$0.80 were removed, but neither dataset includes independent EOG recordings. Residual ocular or muscle artifacts therefore cannot be ruled out. We now state this limitation explicitly and interpret the results as discrimination of protocol-defined conditions, not as evidence of decoding cortical attention networks.

Comment 11

Finally, the absence of any discussion regarding model explainability or feature importance is a missed opportunity. Even with a simpler model like XGBoost, the authors could have analyzed which spectral bands and channels were most critical for the cross task and cross dataset predictions. Without an analysis of the underlying features driving the decisions, the claim that models capture context dependent features remains purely speculative. State of the art deep models, such as the ones recommended for citation earlier, often include interpretability modules or attention mechanisms that allow researchers to visualize the spatial and temporal features the network relies on. By neglecting this aspect, the authors fail to provide any neuroscientific insights into why the transfers failed, leaving the paper as a purely empirical exercise in applied machine learning with poor experimental controls.

Response:

Thank you for the suggestion. In the revised manuscript, we added the Results subsection, "Performance consistency across decoders," with a figure comparing all models across the research questions. This provides a broader assessment of whether transfer effects are consistent across decoders. We agree that feature-attribution analysis would be valuable, but given the scope of the full revision, we did not include a separate model-specific feature-importance analysis in this version. If the Reviewer considers this essential, we can add it.

Comment 12

In conclusion, the manuscript fails to deliver on its ambitious initial premises. To be considered for publication, it requires a complete overhaul. The authors must replace their obsolete feature extraction and gradient boosting pipeline with a suite of advanced deep learning architectures, comparing their performance across various transfer scenarios. They must expand their dataset inclusion to encompass a wider variety of cognitive states and tasks, particularly motor imagery and sleep stages, to provide a true measure of paradigm invariance. They must correct the severe semantic errors in their label harmonization process, particularly the grouping of sleep related states with resting states, and fix all grammatical mistakes and punctuation omissions throughout the text. Most importantly, they must implement rigorous, leakage free validation schemes for all their baselines and ensure that cross task evaluations are completely separated from subject identity confounds. Without these major revisions and a significant expansion in scope, the paper remains too superficial and methodologically flawed to contribute meaningfully to the state of the art in electroencephalography decoding.

Response:

We sincerely appreciate this insightful and constructive assessment. We agree that the original manuscript did not sufficiently align its ambitious framing with the scope and rigor of the empirical study. The Reviewer's comments have prompted fundamental revisions throughout the paper.

The manuscript was rewritten as a whole. The Introduction was reformulated around a clearly delimited objective and scope, replacing broad claims about paradigm-invariant EEG decoding with two specific research questions that the available data and experimental design can address rigorously. The Methods were correspondingly expanded: the model suite now covers linear, tree-based and EEG-specific deep architectures, the data-preparation and label-harmonization procedures are described explicitly and corrected, and the validation and cross-task protocols were reworked to remove leakage and subject-identity confounds. The Results section was written anew and is organized around two aggregate figures that answer the two research questions directly, with the interpretation narrowed to what the reported evidence supports. The specific changes, together with the remaining limitations, are documented in the point-by-point responses above, and the manuscript has undergone a comprehensive language and consistency revision.

Overall, these changes have substantially transformed the manuscript in its framing, methodology, evaluation, and presentation. We are grateful for the Reviewer's detailed feedback, which helped us make the study more focused, transparent, and scientifically rigorous. We believe the revised version now delivers on the objectives stated in the Introduction and draws conclusions that are appropriately supported by the reported evidence.